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AN EFFECTIVE IMPLEMENTATION OF FACE RECOGNITION USING DEEP CONVOLUTIONAL NETWORK

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Abstract
Human Face Recognition for forensic investigations and e-governance is widely adopted so that the specific face points can be trained and further investigations can be done. In this approach, the key points of human face with the dynamic features are extracted and trained in the deep neural network model so that the intrinsic aspects of the human face can be realized and further can be used for the criminal investigation or social analytics based applications. In this research manuscript, the usage of deep learning based convolutional network is integrated for the human face analytics and recognition for diversified applications. It is done to have the cavernous evaluation patterns in multiple domains for the knowledge discovery and predictive features of the human face identification domain.

Keywords: face recognition; convolutional neural network; face smile detection.

摘要：广泛用于法医调查和电子政务的人脸识别，以便可以训练特定的人脸点并进行进一步的调查。 在这种方法中，具有动态特征的人脸关键点在深度神经网络模型中被提取和训练，从而可以实现人脸的内在方面，并且可以进一步用于基于犯罪调查或基于社会分析的应用 。 在此研究手稿中，基于深度学习的卷积网络的使用已集成到人脸分析和多种应用程序的识别中。 在人脸识别领域的知识发现和预测特征方面，要在多个领域都具有海绵状的评估模式。

关键词: 人脸识别 卷积神经网络 笑脸检测

I. INTRODUCTION

Nowadays face recognition is widely used as the prime biometric feature in the human face. Face recognition, acknowledgment and distinguishing is the proof of explicit approach with the one of the key research domains for grouped areas including legal applications whereby the suspicious people or articles can be recognized utilizing their live highlights, conduct and attributes [1], [2]. There are numerous sections in the human face which can be prepared and further broke down for the acknowledgment in measurable applications. These items are lips, temple, cheeks, jaw and numerous others which generally make the human grin and push forward to the face grin identification [3]-[9]. In different regions of research, the work on haircut should be possible; however these can be controlled in this manner: the work on face grin recognition is extremely unmistakable. In this exploration work, the profound learning based methodology of Convolutional Neural Network (CNN) with the fuzzy logic is introduced, so the higher level of exactness in the face grin should be possible [10]-[12].

II. AIM OF THE STUDY

The presented aim in the domain of face recognition using deep Convolutional networks is giving the higher degree of accuracy and performance on assorted directions. The displayed result gives the perspective on the

various focuses in the highlights of human face which are very significant and principal to ponder the numerous points of view of the human face

III. HUMAN FACE RECOGNITION

Following are the steps associated with the strategies of human face recognition and it can be integrated with the CNN based deep learning model for the predictive analysis [13]-[15].

a. Parameters
   - Radius
   - Neighbors
   - Grid X
   - Grid Y

b. Training the Algorithm
c. Applying the operation
d. Performing the face recognition:

$$C_m = \begin{cases} 1, & \sum_{i=0}^{I_m-1} F_{m,i} > \theta_m \\ 0, & \text{otherwise} \end{cases}$$

$$F_{m,i} = \begin{cases} \alpha_{m,i}, & \text{if } f_{m,i} > t_{m,i} \\ \beta_{m,i}, & \text{otherwise} \end{cases}$$

If there should be an occurrence of adequacy situated face acknowledgment, the key highlights in powerful arrangement from numerous points of face are removed. Likewise, various directions of the face are extricated from 360 degree viewpoints with the goal that the general assessment of the face should be possible [16]-[18].

The introduced condition and numerical model is broadly related for the face acknowledgment and what is more can be utilized for examination of face grin. The displayed result gives the perspective on the various focuses in the highlights of human face which are very significant and principal to ponder the numerous points of view of the human face and grin [19], [20].

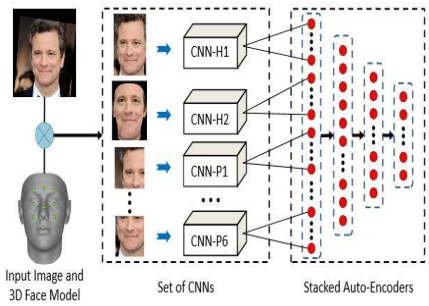

Figure 1. Dynamic Features in Human Face.

Figure 1 is presenting the usage patterns and implementation of human face recognition and thereby the stacked auto encoders so that the deep Convolutional network can be generated.

IV. DEEP CONVOLUTIONAL NEURAL NETWORKS FOR FACE RECOGNITION

Deep Learning is one of the prime research domains in which the exactness to 100% can be accomplished without error factor. The deep learning utilizes Convolutional neural systems in which every neuron and shrouded layer are related with the different superior algorithmic approach for the exactness and least error factor [21], [22].

The preparation procedure utilizes the key pixel positions and the force esteems with the required variables of the human face, therefore, the real impressions of face can be examined [23]-[25].

V. PROJECTED STEPS OF APPROACH EXTRACTION OF PARAMETERS AND KEY FEATURES

In this step, the orientation and dimensions of the digital image of face are analyzed including Radius, Neighboring Pixels, Grid X and Grid Y so that the multiple aspects of the images can be extracted as in Figure 2 and Figure 3. The extraction of features form the key base for the deep neural networks is discussed in [26].

Algorithmic Model Training

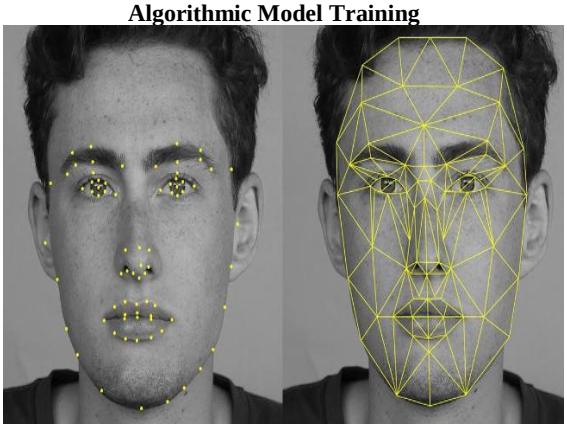

Figure 2. Key Features from a human face

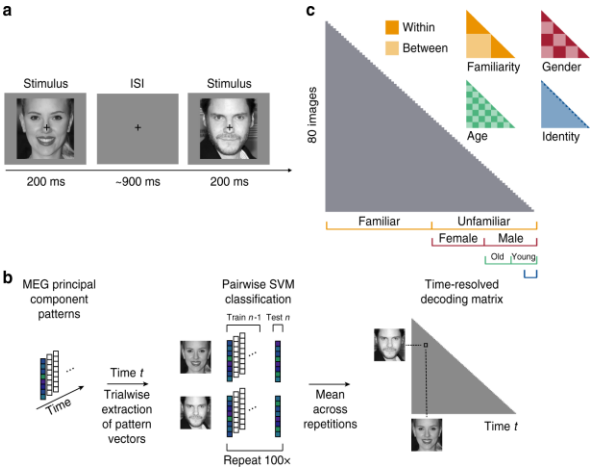

Figure 3. Threshold and Binary Analysis.

VI. EXTRACTION OF THE HISTOGRAMS

The histograms of the image with the smile operations and perspectives are fetched to enable the deep and cavernous analysis of the smile for training as in Figure 4, so that the smile or similar features can be trained.

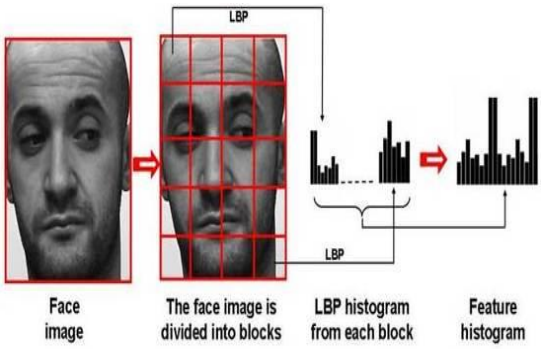

Figure 4. Histogram Extraction and Plotting

Figure 4 presents the extraction and plotting of a histogram obtained from the human face for recognition factors.

VII. PERFORMING CNN WITH FUZZY

The Convolutional neural networks with the fuzzy logic are integrated, so that the accuracy and performance in the analytics phase can be done with the objective function.

$$D = \sqrt{\sum_{i=1}^{n} (hist1_i - hist2_i)^2}$$

In addition, the accuracy level and the percentage can be extracted with the level of matching with the image in the face smile.

Figure 5 depicts the accuracy and identification of the faces using the mathematical formulations with the help of the presented approach.

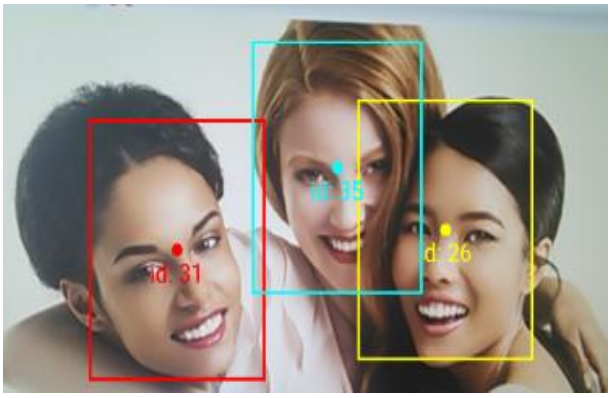

Figure 5. Prediction with the Accuracy and Identification

Figure 6 shows the key demarcation points and extraction patterns of deep learning and machine learning. In case of deep learning, the features points are automatically extracted from the algorithmic approach.

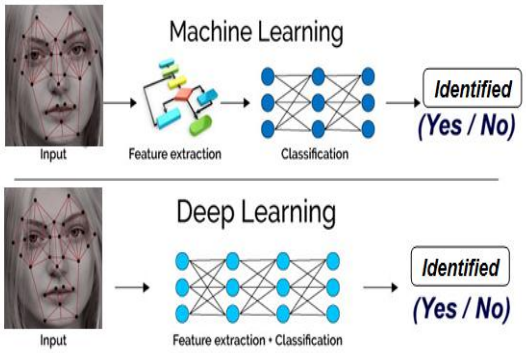

Figure 6. Key Demarcation Points in Deep Learning and Machine Learning

Figure 7 demonstrates the face analytics with the diversified aspects of the Convolutional neural network with the integration of hidden layers for the training and validations.

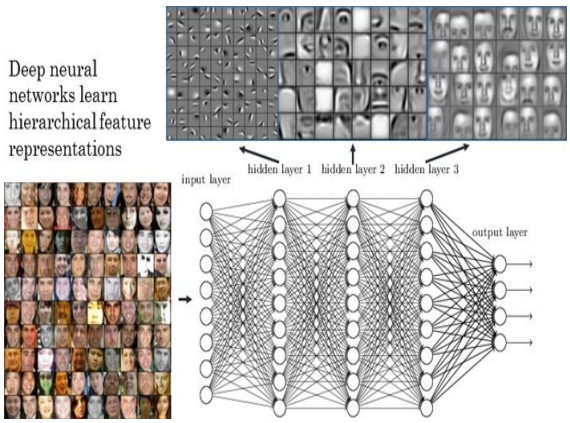

Figure 7. Face Analysis with Assorted Perspectives of CNN

```
importDeep Learning Library
importnumpy as np
fromDeep Learning Library.applications import ModelX16, Model1_v3, Model250, mobilenet

#Load the MODELX model
ModelX_model = ModelX16.MODELX16(weights='imagenet')

#Load the Model1_V3 model
Model1_model = Model1_v3.Model1V3(weights='imagenet')

#Load the Model250 model
Model2_model = Model250.Model250(weights='imagenet')

#Load the MobileNet model
mobilenet_model = mobilenet.MobileNet(weights='imagenet')
fromgoogle.colab import drive
drive.mount('/content/gdrive')
!cd /content/gdrive/My\ Drive
!pwd
!lsgdrive/My\ Drive/
fromDeep Learning Library.preprocessing.image import load_img
fromDeep Learning Library.preprocessing.image import img_to_array
fromDeep Learning Library.applications.imagenet_utils import decode_predictions
importmatplotlib.pyplot as plt
%matplotlib inline

!pwd
!ls
fromgoogle.colab import files
filename = files.upload()
!ls
filename='dog.jpg'
original = load_img(filename, target_size=(224, 224))
print('PIL image size',original.size)
plt.imshow(original)
plt.show()
numpy_image = img_to_array(original)
plt.imshow(np.uint8(numpy_image))
plt.show()
print('numpy array size',numpy_image.shape)
# Convert the image / images into batch format
# expand_dims will add an extra dimension to the data at a particular axis
# We want the input matrix to the network to be of the form (batchsize, height, width, channels)
# Thus we add the extra dimension to the axis 0.
image_batch = np.expand_dims(numpy_image, axis=0)
print('image batch size', image_batch.shape)
plt.imshow(np.uint8(image_batch[0]))
# prepare the image for the MODELX model
processed_image = ModelX16.preprocess_input(image_batch.copy()
)
# get the predicted probabilities for each class
predictions = ModelX_model.predict(processed_image)
# print predictions
# convert the probabilities to class labels
# We will get top 5 predictions which is the default
label = decode_predictions(predictions)
print (label)
```

In the event of AI and deep learning based approach, there is significant outline. On account of CNN and fuzzy based deep learning, there is no compelling reason to unequivocally extricate the highlights. The deep learning coordinated CNN consequently separates the highlights so the general forecast will be increasingly precise with least degree of error factor [23, 24]. Table 1 illustrates this.

Table 1. Comparison between the use of the deep convolution approach and heuristics approach.

	Accuracy (Percentage)	Error Factor (Percentage)	Execution Time (ms)
Heuristics Based Approach	78.39	21.61	9.1282
Deep Convolutional Approach	98.12	1.88	5.8381

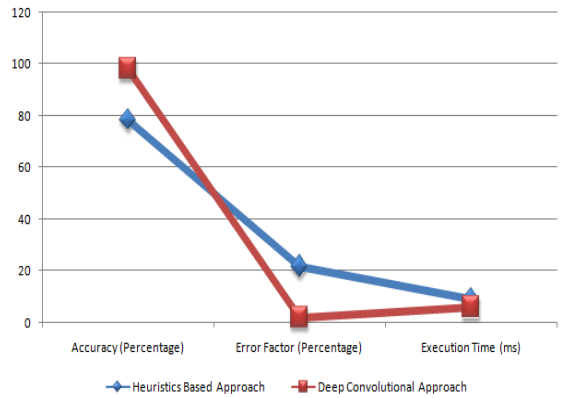

Figure 8. Results Analysis Graph.

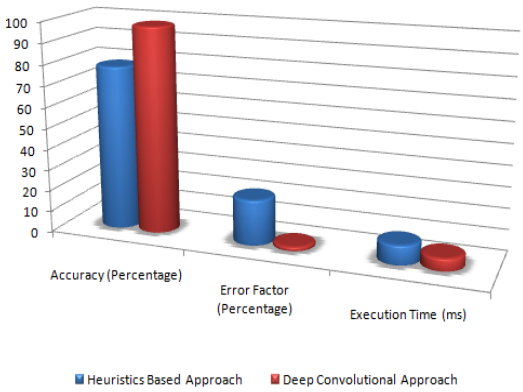

Figure 9: Analysis Graph of Results.

Figures 8 and 9 depict the analysis graphs in the form of bar and line plot which demonstrate that the CNN-based deep learning approach is highly effective and performance aware.

Furthermore, the grin reaction can be assessed with the level of coordinating with the feelings or suppositions of the individual under assessment. The assessment components can be effectively anticipated with the examination of the grin with the reconciliation of higher level of exactness in the CNN-coordinated fuzzy approach.

VIII. Results and Discussion

The domain of face detection is quite effective for the analytics on the assorted factor and is covered here. The results using CNN are very effective with the usage of fine tuning of the pre trained models. Human Face Recognition for criminological examinations and e-administration is broadly accepted with the goal of preparing the particular face focuses and enabling further examinations. In this methodology, the key purposes of human face with the dynamic highlights are separated and prepared in the profound neural system model, so the inborn parts of the human face can be acknowledged and further can be utilized for the criminal examination or social investigation based applications. Furthermore, the face reaction can be assessed with the level of coordinating with the feelings or suppositions of the individual under assessment. The assessment components can be effectively anticipated in case of grin examination with the reconciliation of higher level of exactness in the CNN coordinated fuzzy approach. In this research work, the key points are to introduce the use of deep learning based on Convolutional Neural Network and coordinated for the human face investigation and acknowledgment for expanded applications. It is done to have the enormous assessment designs in various areas for the learning disclosure and prescient highlights of the human face distinguishing proof space. The introduced point in the area of face recognition utilizing deep CNN systems is giving the higher level of precision and execution on the arranged headings. The based approach integrating fuzzy logic with the CNN is giving the effective results and is covered herewith in the manuscript.

IX. Conclusion

To present the cavernous evaluation on the base of biometrics, the usage of human face is widely integrated nowadays. In this approach, the human face is identified from multiple angles and dimensions including face, smile, lips, cheeks, eye brows and many others. These are the key properties of the human face. Using the training data of these features the human face can be further identified and used in the deep learning based approach as described in this manuscript to have the cavernous evaluation and on assorted aspects.

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